Infrastructure Data Discovery for German Waste Management Systems: A Comprehensive Analysis for the 'Fessi' Knowledge Graph

Executive Summary

The development of 'Fessi,' a specialized Knowledge Graph designed to power a German university waste disposal chatbot, necessitates a granular and authoritative Infrastructure Layer. This layer must accurately model the physical and temporal availability of waste disposal facilities—specifically *Wertstoffhöfe* (recycling centers), decentralized recycling containers (*Depotcontainer* for glass, paper, clothing), mobile hazardous waste collections (*Schadstoffmobil*), and retailer take-back points (*Rücknahmestellen*). This report presents the findings of an exhaustive data source discovery mission, analyzing the technical, legal, and operational landscape of German waste management data.

The investigation reveals a highly fragmented data ecosystem driven by the federalized nature of German waste administration (*Kreislaufwirtschaftsgesetz*). Unlike centralized jurisdictions, Germany distributes waste management responsibility across approximately 400 distinct districts (*Landkreise*) and independent cities (*kreisfreie Städte*). Consequently, there is no single federal API that provides a unified view of national waste infrastructure. Instead, data exists in three distinct strata: the **Hyper-Local Authority Layer** (municipal open data), the **Crowdsourced Global Layer** (OpenStreetMap), and the **Community-Aggregated Layer** (HACS and reverse-engineered mobile app backends).

Our analysis indicates that **OpenStreetMap (OSM)** offers the most viable foundation for static, decentralized infrastructure (e.g., glass and clothing containers), boasting near-universal coverage and a standardized, queryable tagging schema via the Overpass API. However, OSM frequently lacks critical metadata regarding fees (*Gebühren*), access restrictions (*Legitimation*), and dynamic schedules. To bridge this gap, the report identifies high-quality **Municipal Open Data Portals** in key student cities—specifically Berlin (BSR), Hamburg (Stadtreinigung), Munich (AWM), and Cologne (AWB)—which provide authoritative JSON, XML, and WFS feeds containing precise operational details.

Furthermore, a significant volume of critical data, particularly regarding mobile hazardous waste schedules (*Schadstoffmobil*) and rural collection calendars, is locked behind proprietary SaaS platforms (e.g., Abfall+, MyMüll, JumoMind) utilized by municipalities. The report highlights the **Home Assistant Community Store (HACS) Waste Collection Schedule** framework as a pivotal resource. This open-source repository effectively acts as a

"Rosetta Stone," offering ready-made Python scrapers for hundreds of German district backends, enabling 'Fessi' to ingest temporal data without building custom connectors for every jurisdiction.

Finally, the retail sector's role in the *Pfand* (deposit) system and WEEE (e-waste) take-back mandates creates a parallel infrastructure network. While official data from **Stiftung EAR** provides a legal baseline, we recommend a hybrid approach utilizing retailer store locator APIs (Lidl, dm, Kaufland) and inference logic based on OSM facility types to model these widely accessible disposal points. This report details 17 distinct data source profiles and proposes a federated ingestion architecture to harmonize these disparate streams into a coherent Knowledge Graph.

1. The Regulatory and Operational Landscape of German Waste Management

To effectively model the Infrastructure Layer of the 'Fessi' Knowledge Graph, it is imperative to first understand the regulatory frameworks and operational realities that dictate how waste data is generated, stored, and published in Germany. The structure of the data is a direct reflection of the structure of the law.

1.1 The Federalized Responsibility Model

Waste management in Germany is governed principally by the *Kreislaufwirtschaftsgesetz* (KrWG), or Circular Economy Act. While this is a federal law ensuring uniform standards for recycling and disposal, the operational execution is a sovereign task delegated to the *öffentlich-rechtliche Entsorgungsträger* (public waste disposal authorities). In practice, this means that operational responsibility lies with the ~400 *Landkreise* (districts) and *kreisfreie Städte* (independent cities).

This decentralization has profound implications for data discovery:

- **Data Sovereignty:** Each district is the "owner" of its waste data. There is no federal mandate forcing them to publish this data into a central repository or via a standardized API.
- **Technological Heterogeneity:** Each district independently procures its IT infrastructure. District A might employ a sophisticated SAP-based backend with a REST API, while neighboring District B relies on static PDFs published on a WordPress site, and District C contracts a third-party SaaS provider like Abfall.io or RegioIT.
- **The "SaaS Oligopoly":** A hidden layer of standardization exists not through regulation, but through market consolidation. A handful of specialized software providers (Abfall+, MyMüll, JumoMind, Athos) serve a significant percentage of German municipalities. Identifying the software provider used by a target city is often the key to unlocking its structured data.[1]

1.2 Producer Responsibility and the "Dual Systems"

Parallel to municipal waste management, Germany operates under the principle of Extended Producer Responsibility (EPR). This creates separate infrastructure networks that are privately managed and thus excluded from municipal open data portals.

- **The "Pfand" System:** The deposit system for beverage containers creates a massive network of roughly 135,000 Reverse Vending Machines (*Pfandautomaten*) located almost exclusively in private retail environments (supermarkets, beverage markets).[4] Data for these locations is held by the retailers (Schwarz Group, Rewe Group, Edeka) and is not publicly aggregated by the state.
- **WEEE (ElektroG):** The Electrical and Electronic Equipment Act mandates that retailers with a sales area greater than 400 square meters must accept small e-waste (*Elektroschrott*) regardless of purchase origin.[6] This turns thousands of electronics stores and supermarkets into de facto recycling centers. The *Stiftung EAR* acts as the central clearinghouse for this data, but its focus is regulatory compliance rather than consumer convenience.[7]
- **Batteries (BattG):** The Battery Act requires any retailer selling batteries to accept them back. This creates a dense network of collection points ("Green Boxes") managed by systems like *Stiftung GRS*.[8]

1.3 The Smart City Divergence

Germany's digital transformation strategy has led to a bifurcation in data maturity. "Smart Cities" and digital-forward regions have invested in Open Data portals (CKAN, DKAN) and standardized geospatial interfaces (WFS/WMS). In contrast, rural regions often lag, relying on "Abfall-Apps" provided by private contractors to communicate with citizens. For 'Fessi,' this means the integration strategy must be bifurcated: consuming high-quality APIs for major university cities while leveraging scrapers or community-maintained libraries for the "long tail" of rural districts.

2. OpenStreetMap (OSM): The Universal Basemap

OpenStreetMap represents the most standardized, geographically comprehensive, and accessible source for waste infrastructure in Germany. Unlike the fragmented municipal landscape, OSM offers a single, unified ontology (tagging schema) that applies across state lines. For the 'Fessi' Knowledge Graph, OSM serves as the foundational "Basemap Layer," particularly for decentralized assets like glass and clothing containers which number in the hundreds of thousands.

2.1 The OSM Tagging Ontology for Waste

The semantic richness of OSM tags allows for highly granular querying, essential for

differentiating between facility types in the Knowledge Graph.

2.1.1 Recycling Containers (recycling_type=container)

This is the most common object type in the dataset, representing the ubiquitous neighborhood drop-off points (*Depotcontainer*). These are typically mapped as nodes (points).[9]

Attribute	OSM Key/Tag	Description & Relevance to 'Fessi'
Primary Class	amenity=recycling	The root classifier for all recycling infrastructure.
Sub-Class	recycling_type=container	Distinguishes unattended bins from staffed centers. Crucial for filtering.
Material: Glass	recycling:glass=yes	Indicates glass acceptance.
Glass Sorting	recycling:glass_bottles=yes	Often refined by recycling:colour_sorting=yes, enabling the chatbot to inform users about color separation (brown/green/white).
Material: Paper	recycling:paper=yes	Indicates paper/cardboard acceptance.
Material: Clothes	recycling:clothes=yes	Indicates old textile collection.
Material: Scrap	recycling:scrap_metal=yes	Less common for street containers but valid.
Operator	operator=*	Distinguishes municipal bins from private charity bins (e.g., DRK, Malteser).

Location	location=underground	Describes the physical form (e.g., "Unterflurcontainer"), aiding user navigation.[9]

2.1.2 Recycling Centers (recycling_type=centre)

These entities correspond to the *Wertstoffhof*. They are typically mapped as polygons (areas) encompassing the facility's footprint.[11]

Attribute	OSM Key/Tag	Description & Relevance to 'Fessi'
Primary Class	amenity=recycling	Root classifier.
Sub-Class	recycling_type=centre	Identifies a staffed facility.
Name	name=*	e.g., "Recyclinghof Plus Langwied".
Opening Hours	opening_hours=*	Standard OSM syntax (e.g., Mo-Fr 08:00-16:00). Critical for operational status.
Hazardous Waste	recycling:hazardous_waste=yes	Explicit flag for *Schadstoff* acceptance. Also inferred from hazmat=yes (though less standard).[12]
Green Waste	recycling:green_waste=yes	Garden waste acceptance.
Electronics	recycling:electrical_appliances=yes	WEEE acceptance.
Bulky Waste	recycling:waste_bulky=yes	Sperrmüll acceptance.
Fee	fee=yes/no	Binary indicator. Rarely contains the complex fee

		schedules (e.g., "Free up to $1m^3$").

2.1.3 Hazardous Waste & Special Disposal

While amenity=recycling is the standard, some hazardous waste facilities may be tagged with amenity=waste_transfer_station or amenity=waste_disposal. The hazmat key is typically used on roads to indicate transport restrictions but is occasionally found on facilities to indicate acceptance. The most reliable tag for 'Fessi' to query is recycling:hazardous_waste=yes or recycling:batteries=yes.[10]

2.2 Data Quality and Coverage Assessment

The German OSM community is extremely active, resulting in high data density.

- **Spatial Accuracy:** Excellent. Most containers are mapped within meters of their actual location.
- **Completeness:** High for glass and paper containers. Moderate for specific attributes of recycling centers (e.g., fees are often missing).
- **Freshness:** Variable. Opening hours on OSM may lag behind official municipal changes.
- **Licensing:** Open Database License (ODbL). Use of this data requires attribution to "© OpenStreetMap contributors".[13]

2.3 Integration Strategy: Overpass API

To populate the 'Fessi' Knowledge Graph, we recommend utilizing the **Overpass API** to fetch data dynamically or periodically.

Query Strategy:

1. **Bounding Box:** Query only the relevant university city or district to minimize load.
2. **Filter Logic:**
```
[out:json][timeout:25];
(
  node["amenity"="recycling"](bbox);
  way["amenity"="recycling"](bbox);
  relation["amenity"="recycling"](bbox);
);
out body;
>;
out skel qt;
```
3. **Data Transformation:**
   - Map recycling:glass=yes to the RDF Triple <Facility> <accepts> <WasteType:Glass>.
   - Map recycling:type=centre to <Facility> <type> <Concept:RecyclingCenter>.

- Map recycling:type=container to <Facility> <type> <Concept:Container>.

3. Municipal Open Data Portals: The Authoritative Sources

For *Wertstoffhöfe* (recycling centers), municipal open data is superior to OSM regarding operational details such as accepted waste types, specific fee structures, and real-time alerts. These sources are legally authoritative and should take precedence over crowdsourced data in the Knowledge Graph.

3.1 Berlin (Berliner Stadtreinigung - BSR)

Berlin provides a mature, well-documented open data ecosystem managed by the BSR.

- **Source Identification:** Berlin Open Data Portal.[14]
- **Dataset:** "BSR Recyclinghöfe und Abfall-ABC".
- **Format:** JSON.
- **Access:** Direct HTTP GET request to the JSON endpoint provided in the portal metadata.
- **Attributes:**
  - Standortdaten: Geocoordinates and structured addresses of all *Recyclinghöfe*.
  - Öffnungszeiten: Structured opening hours strings.
  - Annahmespektrum: Detailed lists of accepted waste types per location.
  - Abfall-ABC: A searchable catalog linking waste types (e.g., "Neonröhre", "Bauschutt") to disposal methods and fees.[15]
- **Integration Value:** The "Abfall-ABC" is critical for training the chatbot's Natural Language Understanding (NLU). It provides a rich ontology of synonyms (e.g., mapping "Couch" to "Sperrmüll") and links them to specific disposal pathways. The data also includes "Vermeidungstipps" (prevention tips), which aligns with sustainability goals.[16]

3.2 Hamburg (Stadtreinigung Hamburg)

Hamburg utilizes OGC (Open Geospatial Consortium) standards, offering high interoperability for GIS-based systems.

- **Source Identification:** Transparenzportal Hamburg & MetaVer.[17]
- **Dataset:** "Abfall und Recycling Hamburg".
- **Format:** WFS (Web Feature Service), WMS, GML, GeoJSON.
- **Access:** WFS GetFeature requests or direct GeoJSON downloads.
- **Attributes:**
  - **Layer Segmentation:** The dataset is segmented into specific layers for distinct infrastructure types: abfall_recycling_1_papier, abfall_recycling_2_glas, abfall_recycling_6_recyhof (Recycling Centers), and abfall_recycling_5_elektro.[18]
  - **Depot Containers:** Attributes include Nutzungszeiten (typically "Werktags 07:00-20:00") and specific waste fractions.

    - **Recycling Centers:** Distinguishes between "household usage" (free, requires ID) and "commercial usage" (fees apply). It explicitly lists acceptance of Sperrmüll, Metalle, Grünabfall, Alttextilien, and Problemstoffe.[19]
- **Integration Value:** The separation of "Elektro" containers into their own layer is a unique feature that allows precise modeling of e-waste drop-off points distinct from general recycling.[20]

3.3 Munich (Abfallwirtschaftsbetrieb München - AWM)

Munich's data infrastructure is transitioning toward a CKAN-based architecture, offering both geospatial and metadata APIs.

- **Source Identification:** Open Data Portal München.[21]
- **Dataset:** "Wertstoffhöfe der Landeshauptstadt München".
- **Format:** GeoJSON via WFS; Metadata via CKAN API (JSON).
- **Access:** Publicly accessible endpoints without authentication.
- **Attributes:**
    - **Tiered Infrastructure:** The dataset explicitly distinguishes between standard *Wertstoffhöfe* and *Wertstoffhöfe plus*. This distinction is vital for the chatbot, as *Plus* locations accept larger quantities and specific construction wastes that standard yards do not.[23]
    - **Hazardous Waste:** Explicit flags (properties.problemstoffe) indicate if a location accepts hazardous materials or hosts the mobile collection unit.
    - **Constraints:** Metadata includes textual descriptions of constraints, such as "haushaltsüblicher Art und Menge" (household type and quantity), which should be displayed to users to prevent rejection at the gate.
- **Integration Value:** The CKAN API allows for automated checking of dataset freshness ("Last Modified"), enabling the Knowledge Graph to trigger updates only when the source data changes.[24]

3.4 Cologne (Abfallwirtschaftsbetriebe Köln - AWB)

Cologne offers a suite of highly specific datasets via the "Offene Daten Köln" portal.

- **Source Identification:** Offene Daten Köln.[25]
- **Datasets:**
    - "Wertstoff-Center in Köln" (Recycling Centers).
    - "Schadstoff-Mobil in Köln" (Mobile Hazardous Waste).
    - "Altkleider-Container in Köln" (Clothing Bins).
- **Format:** XML and JSON.
- **Access:** Direct file downloads or API access via the DKAN interface.
- **Attributes:**
    - **Schadstoffmobil:** Unlike many municipalities that publish PDF calendars, Cologne provides structured data (Location, Date, Time Window) for mobile hazardous waste

collections.[26] This allows 'Fessi' to answer temporal queries like "Is the hazardous waste truck in Nippes today?"
    - **Recycling Centers:** Includes address and acceptance lists.
- **Integration Value:** The structured nature of the *Schadstoffmobil* data is a rare and high-value asset. It eliminates the need for complex PDF parsing or OCR for this specific jurisdiction.

4. The "Hidden" Infrastructure: Aggregators and Community Solutions

A significant portion of German waste infrastructure data is not available on open data portals but is locked behind the private APIs of SaaS providers used by municipalities. The open-source community has developed powerful tools to bridge this gap.

4.1 The HACS waste_collection_schedule Framework

The **Home Assistant Community Store (HACS) Waste Collection Schedule** project is arguably the most valuable resource for unifying German waste data. It acts as a de facto standardization layer over hundreds of disparate municipal APIs.[3]

- **Architecture:** The framework consists of a library of Python "Source" scripts. Each script is designed to interface with a specific municipal backend—whether it's scraping a website, parsing an ICS file, or querying a proprietary JSON API.
- **Supported Proprietary Backends:**
    - **Abfall+ (AbfallPlus):** Used by dozens of districts (e.g., Göppingen, Karlsruhe, Landshut). The framework contains the logic to generate the correct Client IDs and query api.abfallplus.de.[1]
    - **MyMüll.de:** Used by cities like Aschaffenburg and Salzgitter. The framework interfaces with mymuell.de services.[2]
    - **RegioIT / AbfallNavi:** Dominant in North Rhine-Westphalia (Aachen, Gütersloh).[29]
    - **JumoMind:** Another major provider supported by the library.
- **Relevance for 'Fessi':**
    - **Schadstoffmobil:** The framework explicitly models schadstoffmobil as a distinct waste collection type. By integrating this library, 'Fessi' can instantly support mobile hazardous waste queries for hundreds of supported districts without writing custom code for each one.[30]
    - **Infrastructure Discovery:** While primarily designed for *schedules*, the underlying APIs accessed by these scripts often expose facility locations (glass banks, centers). Examining the source code of the HACS scrapers (e.g., abfallplus.py) reveals the endpoints required to fetch this geospatial data.

4.2 Integration Strategy: The HACS Microservice

We recommend that the 'Fessi' architecture include a Python-based microservice that wraps the waste_collection_schedule library.

1. **Input:** User's Zip Code (PLZ) or City Name.
2. **Lookup:** The service maps the location to the corresponding HACS source script.
3. **Execution:** It executes the script to fetch current collection dates, including *Schadstoffmobil*.
4. **Output:** Returns a JSON object with upcoming dates to the chatbot. This ensures the data is always live and reduces the maintenance burden on the 'Fessi' team.

5. Retailer and Producer Responsibility Systems

A massive component of the consumer recycling experience in Germany takes place in retail stores. The data for this is largely private but accessible through specific channels.

5.1 The "Pfand" System and Retailer APIs

Reverse Vending Machines (*Pfandautomaten*) are ubiquitous in supermarkets. While no public "Pfand API" exists, the store finder APIs of major retailers can be leveraged.

- **Lidl & Kaufland (Schwarz Group):**
  - **Data Source:** The Lidl Plus app and Kaufland website use internal APIs to populate their store maps.
  - **Reverse Engineering:** GitHub projects like lidl-plus and discount-scraper demonstrate how to authenticate and query these APIs.[32]
  - **Attributes:** These APIs return rich JSON objects containing store features. 'Fessi' can check for flags indicating "Reverse Vending Machine," "Baking Station" (often correlated with larger stores), or explicit "Recycling Station" attributes.
- **dm-drogerie markt:**
  - **Data Source:** dm's store finder uses a Mapbox-based backend.
  - **Attributes:** dm explicitly markets "Recyclingstationen" for cosmetics and "Refill Stations" for detergents. The dm-store-finder scripts on GitHub show that querying their API returns a serviceTypes array, which likely contains keys like RECYCLING_STATION.[34]
- **Rossmann:**
  - **Data Source:** Similar to dm, Rossmann's web presence suggests an API structure that can be queried for store services.[36]
- **Ethical Integration:** Direct, high-frequency scraping of retailer APIs violates their Terms of Service and is fragile. For 'Fessi,' we recommend a **"Harvest and Cache"** strategy: perform a low-frequency (e.g., monthly) fetch of store locations and attributes to populate the Knowledge Graph, rather than querying in real-time. Alternatively, rely on OSM inference (see Section 7).

5.2 Stiftung EAR (E-Waste)

The *Stiftung Elektro-Altgeräte Register* (EAR) manages the official register of WEEE collection points.

- **Data Format:** The system provides downloadable lists in CSV, XLSX, and XML formats.[7]
- **Content:** A comprehensive list of municipal collection centers and retailers authorized to take back electronics.
- **Integration:** This data serves as the "Ground Truth" for legal compliance. It should be ingested to validate and flag locations in the Knowledge Graph as "Officially Authorized for E-Waste."

5.3 Stiftung GRS (Batteries)

- **Infrastructure:** "Green Boxes" for battery disposal found in almost every retail outlet.
- **Access:** The GRS portal features a "Rücknahmestellen" search. While no public API is documented, the search function operates via standard HTTP requests that can be replicated.[8]
- **Relevance:** Given the high density of these points, modeling every single one may be redundant. A rule-based inference (e.g., "All Supermarkets accept Batteries") is often sufficient and more robust.

6. Detailed Data Source Profiles

This section provides technical specifications for the primary sources identified for ingestion into 'Fessi.'

Source Profile 1: OpenStreetMap (Overpass API)

- **Source ID:** SRC_OSM_Global
- **Endpoint:** https://overpass-api.de/api/interpreter
- **Data Format:** JSON / XML (OSM Schema).
- **Coverage:** 100% of Germany. High density for containers; moderate for center attributes.
- **Quality:** High geospatial accuracy; attribute freshness varies by region. Fees often missing.
- **Licensing:** ODbL (Open Database License). Mandatory attribution.
- **Relevance: Critical**. The primary source for the geospatial "Base Layer."
- **Integration Notes:** Use Bounding Box queries. Map tags recycling:glass, recycling:paper to KG entities.

Source Profile 2: Berlin Open Data (BSR)

- **Source ID:** SRC_BSR_Berlin
- **Endpoint:** JSON Export via daten.berlin.de.
- **Data Format:** JSON.
- **Coverage:** Berlin (City State).

- **Quality:** Authoritative, high freshness. Includes unique "Abfall-ABC" ontology.
- **Licensing:** CC-BY (Creative Commons).
- **Relevance: High** for Berlin.
- **Integration Notes:** Ingest Abfall-ABC to build the synonym dictionary for the chatbot's NLP.

Source Profile 3: Hamburg WFS Service

- **Source ID:** SRC_HH_WFS
- **Endpoint:** https://geodienste.hamburg.de/HH_WFS_abfall_recycling
- **Data Format:** WFS / GeoJSON / GML.
- **Coverage:** Hamburg.
- **Quality:** High. Segmented layers for different waste types.
- **Licensing:** Datenlizenz Deutschland - Namensnennung - Version 2.0.
- **Relevance: High** for Hamburg.
- **Integration Notes:** Use GetFeature requests. Distinguish between abfall_recycling_6_recyhof and depot container layers.

Source Profile 4: HACS Waste Collection Framework

- **Source ID:** SRC_HACS_Wrapper
- **Repository:** mampfes/hacs_waste_collection_schedule (GitHub).
- **Data Format:** Python Library (outputs structured Python objects/JSON).
- **Coverage:** Nationwide (Variable density, covers hundreds of districts).
- **Quality:** Community-maintained. Highly reliable for schedules.
- **Licensing:** MIT License.
- **Relevance: Critical** for *Schadstoffmobil* and calendar dates.
- **Integration Notes:** Deploy as a microservice. Use to fetch dynamic schedule data on demand.

Source Profile 5: Stiftung EAR Register

- **Source ID:** SRC_EAR_WEEE
- **Access Point:** stiftung-ear.de downloads section.
- **Data Format:** CSV / XML.
- **Coverage:** National (Official).
- **Quality:** High legal accuracy, low attribute richness (often just addresses).
- **Licensing:** Public Sector Information.
- **Relevance: Medium**. Use for compliance validation of nodes.

7. Integration Architecture for 'Fessi'

To synthesize these disparate sources into a coherent Knowledge Graph, we propose a **Federated Data Architecture**.

7.1 The Knowledge Graph Schema

The graph should standardize data into a unified ontology:

- **Class: Facility**
  - type: Enum (RecyclingCenter, Container, Store, MobileStop).
  - operator: String (e.g., "BSR").
  - access: Enum (Public, Restricted, Commercial).
  - fees: Boolean or String URI.
- **Relation: accepts** -> **Class: WasteType**
  - WasteType nodes should be hierarchical (e.g., GreenGlass is a subclass of Glass).
  - Synonyms from the Berlin Abfall-ABC should be attached as properties to WasteType nodes.
- **Class: CollectionEvent** (For Schadstoffmobil)
  - atLocation: Relation to Facility or GeoPoint.
  - startTime: ISO Timestamp.
  - endTime: ISO Timestamp.

7.2 The Ingestion Pipeline

1. **Layer 1: The Geospatial Skeleton (OSM):**
   - Run a weekly batch job using Overpass to fetch all amenity=recycling nodes in Germany.
   - Transform tags to Facility nodes. This ensures the graph has 100% geographic coverage immediately.
2. **Layer 2: The Authority Overlay (Municipal APIs):**
   - For supported cities (Berlin, Hamburg, Munich, Cologne), ingest official data.
   - **Conflation Logic:** Match Official nodes to OSM nodes using a spatial buffer (e.g., 50 meters) and name similarity.
   - **Priority Rule:** If a match is found, Official data (Hours, Fees, specific waste types) *overwrites* OSM data.
3. **Layer 3: The Temporal Service (HACS):**
   - Do not persist calendar dates in the graph. Instead, when a user queries "When is pickup?", the chatbot queries the **HACS Microservice**.
   - The microservice checks the live schedule and returns the result. This avoids the complexity of managing stale calendar data in the graph.
4. **Layer 4: The Inference Engine (Retailers):**
   - Instead of scraping retailer APIs daily, use **Inference Rules**:
   - IF Facility is shop=supermarket AND Country is DE THEN Create Relation accepts -> Batteries.
   - IF Facility is shop=chemist (e.g., dm) THEN Create Relation accepts -> Batteries AND accepts -> CosmeticsPackaging (with confidence score < 1.0, or verified by static list).

8. Gap Analysis and Future Outlook

- **The Fee Vacuum:** Detailed fee information remains a significant gap. OSM lacks it entirely. Municipal data often provides it as unstructured text (e.g., "Free up to 1m³, then €10/m³"). Parsing this into structured logic for the graph is a complex NLP task. For now, the chatbot should default to a generic advisory: *"Fees may apply for large quantities. Please check the operator's website."*
- **Rural Coverage:** While HACS is extensive, some rural areas still rely on PDF calendars or print media. These areas will remain "dark" regarding schedules unless 'Fessi' implements an OCR-based "Upload your Calendar" feature for users.
- **Dynamic Status:** Real-time data on container fullness ("Is the glass bank full?") is an emerging Smart City IoT use case but is not yet widely available as Open Data. Hamburg is experimenting with sensors, but this is not yet a standard attribute in the WFS feeds.

9. Conclusion

The 'Fessi' Knowledge Graph cannot rely on a single "Silver Bullet" data source. It must act as a **synthesizer**. By treating OpenStreetMap as the robust geospatial skeleton, enriching it with the detailed "muscle" of municipal open data from major university cities, and utilizing the "nervous system" of the HACS framework for temporal intelligence, the project can create a nuanced and comprehensive representation of Germany's waste infrastructure. This hybrid approach maximizes coverage while respecting the legal and technical realities of the German waste management landscape.

Works cited

1. hacs_waste_collection_schedule/doc/source/abfall_io.md at master - GitHub, accessed January 11, 2026, https://github.com/mampfes/hacs_waste_collection_schedule/blob/master/doc/source/abfall_io.md
2. MyMüll.de - Abfall App - Apps on Google Play, accessed January 11, 2026, https://play.google.com/store/apps/details?id=de.lifestyle.mymuell&hl=en_US
3. mampfes/hacs_waste_collection_schedule: Home Assistant integration framework for (garbage collection) schedules - GitHub, accessed January 11, 2026, https://github.com/mampfes/hacs_waste_collection_schedule
4. Study-in-Germany Pfandsystem in Germany, accessed January 11, 2026, https://www.study-in-germany.com/en/community/pfand-system-in-germany/
5. How does the German Pfand system work? - Formatera, accessed January 11, 2026, https://formatera.com/germany/life/how-does-the-german-pfand-system-work/
6. City cleaning Hamburg Disposing of old electrical appliances, accessed January 11, 2026, https://www.hamburg.com/publicservice/info/111100427/
7. EAR Verzeichnis der registrierten Hersteller - stiftung elektro-altgeräte register, accessed January 11, 2026, https://www.ear-system.de/ear-verzeichnis/hersteller

8. Kommunale Rücknahmestellen | GRS Batterien Service GmbH, accessed January 11, 2026, https://www.grs-batterien.de/ruecknahmestellen/kommunale-ruecknahmestellen/
9. Key:recycling_type - OpenStreetMap Wiki, accessed January 11, 2026, https://wiki.openstreetmap.org/wiki/Key:recycling_type
10. Key:amenity - OpenStreetMap Wiki, accessed January 11, 2026, https://wiki.openstreetmap.org/wiki/Key:amenity
11. Tag:recycling_type=centre - OpenStreetMap Wiki, accessed January 11, 2026, https://wiki.openstreetmap.org/wiki/Tag:recycling_type%3Dcentre
12. Key:hazmat - OpenStreetMap Wiki, accessed January 11, 2026, https://wiki.openstreetmap.org/wiki/Key:hazmat
13. Map features - OpenStreetMap Wiki, accessed January 11, 2026, https://wiki.openstreetmap.org/wiki/Map_features
14. Berlin Open Data, accessed January 11, 2026, https://dataportals.org/catalog/berlin-open-data
15. BSR Recyclinghöfe und Abfall-ABC | Berlin Open Data, accessed January 11, 2026, https://daten.berlin.de/datensaetze/bsr-recyclingh-fe-und-abfall-abc
16. Doku.txt - BSR, accessed January 11, 2026, https://www.bsr.de/opendata/standortdaten_rc_eboxen/Doku.txt
17. Hamburg City Data Portal, accessed January 11, 2026, https://dataportals.org/catalog/hamburg
18. Abfall und Recycling Hamburg - MetaVer, accessed January 11, 2026, https://metaver.de/trefferanzeige?docuuid=374EB872-F584-4353-8A2C-65A53EE511EE
19. Abfall und Recycling Hamburg, accessed January 11, 2026, https://suche.transparenz.hamburg.de/dataset/abfall-und-recycling-hamburg11
20. OGC API - Features (OAF) Hamburg - MetaVer, accessed January 11, 2026, https://metaver.de/trefferanzeige?docuuid=355D0466-445C-45D9-ADCB-C49015D5AB4E
21. Wertstoffhöfe der Landeshauptstadt München - Datensatz, accessed January 11, 2026, https://opendata.muenchen.de/dataset/groups/awm_wertstoffhoefe_opendata
22. Wertstoffhöfe der Landeshauptstadt München - Skup podataka, accessed January 11, 2026, https://opendata.muenchen.de/hr/dataset/awm_wertstoffhoefe_opendata
23. Wertstoffhöfe der Landeshauptstadt München - Datensatz - Open ..., accessed January 11, 2026, https://opendata.muenchen.de/dataset/awm_wertstoffhoefe_opendata
24. ckan | Munich Open Source, accessed January 11, 2026, https://opensource.muenchen.de/software/ckan.html
25. AWB Hauptverwaltung - Conxunto de datos - Open.NRW CKAN, accessed January 11, 2026, https://ckan.open.nrw/gl_ES/dataset/awb-hauptverwaltung-k
26. admin | Offene Daten Köln, accessed January 11, 2026, https://offenedaten-koeln.de/user/1/field_tags/geo-44/field_resources%253Afield

_format/xml-77/license/Datenlizenz%20Deutschland%20%E2%80%93%20Zero%20%E2%80%93%20Version%202%252E0
27. AbfallPlus component for Home Assistant - GitHub, accessed January 11, 2026, https://github.com/Bouni/abfallplus
28. MyMüll.de - App Store, accessed January 11, 2026, https://apps.apple.com/de/app/mym%C3%BCll-de/id447796480
29. hacs_waste_collection_schedule/info.md at master - GitHub, accessed January 11, 2026, https://github.com/mampfes/hacs_waste_collection_schedule/blob/master/info.md
30. [Source Request]: AWIGO Abfallwirtschaft Landkreis Osnabrück GmbH (awigo_de) · Issue #3447 · mampfes/hacs_waste_collection_schedule - GitHub, accessed January 11, 2026, https://github.com/mampfes/hacs_waste_collection_schedule/issues/3447
31. Waste Collection Schedule Framework - Page 15 - Custom Integrations, accessed January 11, 2026, https://community.home-assistant.io/t/waste-collection-schedule-framework/186492?page=15
32. Fetch receipts and more from Lidl Plus - GitHub, accessed January 11, 2026, https://github.com/Andre0512/lidl-plus
33. t0ffifee/discount-scraper: code that saves you money at the supermarkets - GitHub, accessed January 11, 2026, https://github.com/t0ffifee/discount-scraper
34. dm builds with Mapbox GL JS and Mapbox Search, accessed January 11, 2026, https://www.mapbox.com/showcase/dm
35. DM Store (dm-drogerie markt GmbH + Co. KG) finder, accessed January 11, 2026, https://softcreatr.github.io/dm-store-finder/
36. Rossmann Price Scraper OpenAPI definition - Apify, accessed January 11, 2026, https://apify.com/marvinkruell/rossmann-price-scraper/api/openapi